

Differential methylation of OPRK1 in borderline personality disorder is associated with childhood trauma.

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According to a growing body of neurobiological evidence, the core symptoms of borderline personality disorder (BPD) may be linked to an opioidergic imbalance between the hedonic and stimulatory activity of mu opioid receptors (MOR) and the reward system inhibiting effects of kappa opioid receptors (KOR). Childhood trauma (CT), which is etiologically relevant to BPD, is also likely to lead to epigenetic and neurobiological adaptations by extensive activation of the stress and endogenous opioid systems. In this study, we investigated the methylation differences in the promoter of the KOR gene (OPRK1) in subjects with BPD (N = 47) and healthy controls (N = 48). Comparing the average methylation rates of regulatorily relevant subregions (specified regions CGI-1, CGI-2, EH1), we found no differences between BPD and HC. Analyzing individual CG nucleotides (N = 175), we found eight differentially methylated CG sites, all of which were less methylated in BPD, with five showing highly interrelated methylation rates. This differentially methylated region (DMR) was found on the falling slope (5') of the promoter methylation gap, whose effect is enhanced by the DMR hypomethylation in BPD. A dimensional assessment of the correlation between disease severity and DMR methylation rate revealed DMR hypomethylation to be negatively associated with BPD symptom severity (measured by BSL-23). Finally, analyzing the influence of CT on DMR methylation, we found DMR hypomethylation to correlate with physical and emotional neglect in childhood (quantified by CTQ). Thus, the newly identified DMR may be a biomarker of the risks caused by CT, which likely epigenetically contribute to the development of BPD.

Borderline and Depression: A Thin EEG Line.

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In everyday clinical practice, there is an ongoing debate about the nature of major depressive disorder (MDD) in patients with borderline personality disorder (BPD). The underlying research does not give us a clear distinction between those 2 entities, although depression is among the most frequent comorbid diagnosis in borderline personality patients. The notion that depression can be a distinct disorder but also a symptom in other psychopathologies led our team to try and delineate those 2 entities using 146 EEG recordings and machine learning. The utilized algorithms, developed solely for this purpose, could not differentiate those 2 entities, meaning that patients suffering from MDD did not have significantly different EEG in terms of patients diagnosed with MDD and BPD respecting the given data and methods used. By increasing the data set and the spatiotemporal specificity, one could have a more sensitive diagnostic approach when using EEG recordings. To our knowledge, this is the first study that used EEG recordings and advanced machine learning techniques and further confirmed the close interrelationship between those 2 entities.

Human olfactory neural progenitor cells reveal differences in IL-6, IL-8, thrombospondin-1, and MCP-1 in major depression disorder and borderline personality disorder.

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Discovering biological markers is essential for understanding and treating mental disorders. Despite the limitations of current non-invasive methods, neural progenitor cells from the olfactory epithelium (hNPCs-OE) have been emphasized as potential biomarker sources. This study measured soluble factors in these cells in Major Depressive Disorder (MDD), Borderline Personality Disorder (BPD), and healthy controls (HC). We assessed thirty-five participants divided into MDD (n=14), BPD (n=14), and HC (n=7). MDD was assessed using the Hamilton Depression Rating Scale. BPD was evaluated using the DSM-5 criteria and the Structured Clinical Interview for Personality Disorders. We isolated hNPCs-OE, collected intracellular proteins and conditioned medium, and quantified markers and soluble factors, including Interleukin-6, interleukin-8, and others. Analysis was conducted using one-way ANOVA or Kruskal-Wallis test and linear regression. We found that hNPCs-OE of MDD and BPD decreased Sox2 and laminin receptor-67 kDa levels. MASH-1 decreased in BPD, while tubulin beta-III decreased in MDD compared to controls and BPD. Also, we found significant differences in IL-6, IL-8, MCP-1, and thrombospondin-1 levels between controls and MDD, or BPD, but not between MDD and BPD. Altered protein markers are evident in the hNPCs-OE in MDD and BPD patients. These cells also secrete higher concentrations of inflammatory cytokines than HC cells. The results suggest the potential utility of hNPCs-OE as an in vitro model for researching biological protein markers in psychiatric disorders. However, more extensive validation studies are needed to confirm their effectiveness and specificity in neuropsychiatric disorders.

Altered brain activity and childhood trauma in Chinese adolescents with borderline personality disorder.

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Childhood trauma may cause borderline personality disorder (BPD). The aim of this study was to assess functional alteration and its association with childhood trauma in Chinese adolescents with BPD. A total of 187 adolescents with BPD aged 12-17 years and 207 age and gender- matched healthy controls (HCs) were enrolled into this study. The sample consisted of 50 adolescents with BPD and 21 HCs underwent brain resting-state functional magnetic resonance imaging (rs-fMRI). The rs-fMRI data was analyzed for both neural activity as indicated by amplitude of low-frequency fluctuation (ALFF) and seed-based functional connectivity (FC). Clinical assessment for childhood trauma, impulsivity, and depression was also performed. Correlative analysis of functional alterations with childhood trauma assessment were performed. Adolescents with BPD had significantly higher rate of all assessed childhood trauma than the HC group ($P < 0.001$). Most adolescents with BPD (61.5 %) had emotional neglect, which was the most commonly seen type of childhood trauma. Compared with HCs, adolescents with BPD showed decreased ALFF in the cortical regions including the left superior frontal gyrus and right middle occipital gyrus, and default mode network (DMN) regions including the left angular gyrus and medial superior frontal gyrus. Adolescents with BPD also showed enhanced ALFF in the limbic system (left hippocampus, insula, thalamus) ($P < 0.05$, FWE correction, cluster size ≥ 100). There were significant correlations between the insula ALFF and childhood trauma assessment for emotional neglect, physical abuse and physical neglect ($P < 0.01$). Moreover, adolescents with BPD showed increased FC between the left insula and right cortical regions (voxel $P < 0.001$, cluster $P < 0.05$, FWE correction). The sample size was small. This cohort had patients with more severe BPD symptoms and some had comorbidities such as anxiety and obsessive-compulsive disorder. There were alterations of brain activity as indicated by ALFF in the limbic - cortical circuit and DMN regions in adolescents with BPD and the activity in the left insula was correlated with emotional neglect. In addition, the FC between the left insula and the limbic - prefrontal circuit was enhanced. These results implicate that the functional alterations of insula may serve as a potential neuroimaging biomarker for adolescents with BPD who suffered from childhood trauma.

miR-124-3p target genes identify globus pallidus role in suicide ideation recovery in borderline personality disorder.

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Borderline personality disorder (BPD) is characterized by patterns of unstable affect, unstable interpersonal relationships, and chronic suicidal tendencies. Research on the genetics, epigenetics, and brain function of BPD is lacking. MicroRNA-124-3p (miR-124-3p) was recently identified in a Genome-Wide Association Study as likely associated with BPD. Here, we identified the anatomical brain expression of genes likely modulated by miR-124-3p and compared morphometry in those brain regions in BPD inpatients vs. controls matched for psychiatric comorbidities. We isolated lists of targets likely modulated by miR-124-3p from TargetScan (v 8.0) by their preferentially conserved targeting (Aggregate PCT > 0.99, see Supplementary Table 1). We applied Process Genes List (PGL) to identify regions of interest associated with the co-expression of miR-124-3p target genes. We compared the gray matter volume of the top region of interest co-expressing those genes between BPD inpatients (n = 111, 46% female) and psychiatric controls (n = 111, 54% female) at The Menninger Clinic in Houston, Texas. We then correlated personality measures, suicidal ideation intensity, and recovery from suicidal ideation with volumetrics. Gene targets of miR-124-3p were significantly co-expressed in the left Globus Pallidus (GP), which was smaller in BPD than in psychiatric controls. Smaller GP volume was negatively correlated with agreeableness and with recovery from suicidal ideation post-treatment. In BPD, GP volume may be reduced through miR-124-3p regulation and suppression of its target genes. Importantly, we identified that a reduction of the GP in BPD could serve as a potential biomarker for recovery from suicidal ideation.

Neurophysiological biomarkers of response inhibition and the familial risk for borderline personality disorder.

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Understanding of the biological factors that run in families affected with borderline personality disorder (BPD) is limited. The authors investigated the familial aggregation of neurophysiological biomarkers of response inhibition in the first-degree biological relatives of probands with BPD and associations with psychiatric diagnosis and impulsive traits. In the present study, psychiatric diagnoses and impulsive traits were measured in BPD probands (n = 86), psychiatrically affected and non-affected relatives (n = 60) and controls (n = 83). While undergoing neuroimaging using functional near-infrared spectroscopy, prefrontal cortex (PFC) activation was measured during a go/no-go response inhibition task and compared between probands, relatives and controls. Additionally, non-psychiatrically affected relatives and controls were contrasted to examine the potential impact of familial risk for BPD on response inhibition-related PFC activation in the absence of confounding psychiatric morbidity. Probands showed bilateral decreases in PFC activation during response inhibition compared to relatives and controls. Conversely, both affected and non-affected relatives displayed higher activation than controls and probands in left lateral/medial and right medial PFC, although non-affected relatives showed a lesser extent of activation than affected relatives. Probands and controls reporting greater impulsive traits displayed deactivation across the PFC during response inhibition, whereas relatives showed increased activation. In this first family study of neuroimaging biomarkers in BPD, we show that the familial risk for BPD is reflected in activation of the PFC during response inhibition, with lifetime psychiatric diagnosis and higher impulsive traits in relatives associated with larger increases in PFC activity. Higher PFC activity

during response inhibition including among non-affected relatives could reflect a neurophysiological compensatory mechanism.

Narcissistic and Borderline Personality Disorders: Relationship With Oxidative Stress.

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The authors hypothesized that personality disorders characterized by interpersonal hypersensitivity would be associated with an elevated concentration of 8-hydroxy-2'-deoxyguanosine (8-OH-DG), the oxidized form of guanine, and a biomarker of oxidative stress burden. One hundred ninety-five male and female adults underwent semistructured diagnostic interviews, completed questionnaire measures of social cognition and emotional attribution, and had blood drawn for determination of plasma 8-OH-DG. A hierarchical linear regression model revealed that narcissistic and borderline personality disorders predicted 8-OH-DG level independently of the effects of age, gender, recent alcohol and cigarette use, current major depression, and posttraumatic stress disorder. In all subjects, 8-OH-DG level was also correlated with the number of borderline personality disorder symptoms present. Narcissistic and borderline personality disorders predicted oxidative stress burden independently of potentially confounding factors.

Inflammatory and oxidative endophenotypes in borderline personality disorder: A biomarker cluster analysis.

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This study is designed to search for aggrupation of inflammatory/oxidative biomarker alterations in borderline personality disorder (BPD) and their association with phenotypic features. Inflammatory/nitrosative proteins were measures in plasma and peripheral blood mononuclear cells obtained from BPD patients. Patients were assessed on different clinical dimensions of BPD. Oxidative damage was tested by measuring TBARS, nitrites, catalase, GPx and SOD. Protein expression of I κ B α , NF κ B, iNOS, COX-2, PPAR γ , Keap1, NQO1, Nrf2 and α 7nAChR was also determined. Western blot and ELISA were used for measurements and a cluster analysis of inflammatory/oxidative biomarkers alterations was performed to investigate subgroups of patients with similar alterations and its relationship with clinical features of BPD. 69 patients were included in the study. Two inflammatory/nitrosative clusters of patients were found: Cluster 1 patients showed significantly higher levels of GPx, I κ B α , keap1, NQO1, PPAR γ , α 7nAChR and Nrf2 than cluster 2 patients. These patients had significantly longer duration of illness, milder anxiety symptoms and lower prescription of antipsychotic drugs than cluster 2. Two clusters of BPD patients according to the inflammatory/nitrosative profiles were identified. Cluster 1 had increased antioxidant and anti-inflammatory biomarkers and was characterised by greater chronicity of illness but less acute symptomatic severity.

Biologic markers in borderline personality disorder: a review.

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The use of biologic markers in the evaluation of borderline personality disorder (BPD) patients is reviewed. Many patients with Axis II BPD have coexisting Axis I diagnoses of which depression is the most commonly reported. Biologic markers have not aided in the diagnosis of BPD, but some markers, particularly EEG sleep, are not only abnormal in BPD, but also appear to discriminate Axis I depression from other Axis I codiagnoses. Monoamine oxidase, in vitro red blood cell lithium ratio, and P300 auditory evoked potential when used alone or in a combined diagnostic approach, show promise in identifying these codiagnoses as well. Dexamethasone suppression and thyrotropin-releasing hormone tests appear nonspecific in this population. Pharmacologic trials have demonstrated that some BPD patients have good therapeutic response to antipsychotics and tranylcypromine and poor response to alprazolam.

Neurochemical alterations associated with borderline personality disorder.

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In neuroimaging on borderline personality disorder, prior studies focused on the hippocampus and amygdala, as mentioned above. However, no study investigated whether there were neurochemical changes in the patients with borderline personality disorder. Therefore, in the present study, we aimed to investigate neurochemical change of patients diagnosed with borderline disorder and hypothesized that neurochemicals would change in the hippocampus region of these patients. Seventeen patients and the same number of healthy control subjects were analyzed by using a 1.5 Tesla GE Signa Imaging System. N-acetylaspartate (NAA), choline compounds (CHO), and creatine (CRE) values of hippocampal region were measured. The mean NAA/CRE ratio in the hippocampus region was significantly reduced in the patients with borderline personality disorder compared to that of healthy control subjects. In addition, NAA/CHO ratio of the patients with borderline personality disorder was also significantly reduced when compared to that of healthy subjects. There was no difference in the ratio of CHO/CRE. In summary, we present evidence for reduced NAA in the patients with borderline personality disorder.
